THE RELAXATION TIME DOES NOT CONTROL THE NORMAL APPROXIMATION OF STATIONARY ADDITIVE FUNCTIONALS

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Abstract. Let X be a finite irreducible reversible Markov chain in continuous time with relaxation time τ_2, started from its stationary distribution, and let g be a function on its states with mean 0 and variance 1 under that distribution. Aldous and Fill asked whether the Kolmogorov distance between $\int_0^t g(X_u)\,du/(\sigma t^{1/2})$, where σ^2 is the asymptotic variance rate, and the standard normal law is bounded by a function of $\max|g|$ and t/τ_2 alone that tends to 0 as $t/\tau_2 \to \infty$. It is not. For every $b \geq 1$ and $s \geq 2b$ there are a chain on four states with $\tau_2 = 1$ and a function g of mean 0 and variance 1 with $\max|g| = b$ such that $\int_0^s g(X_u)\,du \leq 0$ with probability at least $1 - 4/s$, so at $t = s$ the distance is at least $1/2 - 4/s$ whatever positive constant the integral is divided by.

1. Introduction

Let X be an irreducible Markov chain in continuous time on a finite set I, with generator $Q = (q_{ij})$ and stationary distribution π, and assume that X is reversible: $\pi_i q_{ij} = \pi_j q_{ji}$ for all i, j. Let $0 = \lambda_1 < \lambda_2 \leq \dots$ be the eigenvalues of $-Q$, and let $\tau_2 = 1/\lambda_2$ be the relaxation time. Write $\mathbb{P}_\pi$ for the chain started from π, $\pi(h) = \sum_i \pi_i h(i)$ and $\|g\|_\infty = \max_i |g(i)|$. A function $g\colon I \to \mathbb{R}$ is *standardized* if $\pi(g) = 0$ and $\pi(g^2) = 1$, and for such g we put

$$S_t = \int_0^t g(X_u)\,du.$$

By Proposition 4.29 of Aldous and Fill [1], the limit $\sigma^2 = \lim_{t\to\infty} t^{-1} \operatorname{var}_\pi S_t$ exists, $0 < \sigma^2 \leq 2\tau_2$, and

$$A(t/\tau_2)\, t\sigma^2 \leq \operatorname{var}_\pi S_t \leq t\sigma^2, \qquad A(u) = 1 - u^{-1}(1 - e^{-u}),$$

so the variance of S_t approaches $t\sigma^2$ at a rate that depends on the chain only through t/τ_2. The law of $S_t/(\sigma t^{1/2})$ under $\mathbb{P}_\pi$ tends to the standard normal law, and the Kolmogorov distance is at most $Kt^{-1/2}$ for $t \geq 1$, with a constant K that depends on the chain [1, Proposition 4.51]. Aldous and Fill remark that the usual proofs give no useful indication of how K depends on the chain, and their Open Problem 4.30 asks whether there is some explicit function $\psi(b, s) \to 0$ as $s \to \infty$, not depending on the chain, such that for standardized g and continuous-time chains

$$\sup_x \Big| \mathbb{P}_\pi\Big(\frac{S_t}{\sigma t^{1/2}} \leq x\Big) - \mathbb{P}(Z \leq x) \Big| \leq \psi(\|g\|_\infty, t/\tau_2), \tag{1.1}$$

where Z is a standard normal random variable. The answer is no.

Theorem 1. *For every $b \geq 1$ and $s \geq 2b$ there are an irreducible reversible chain X on four states with $\tau_2 = 1$ and a standardized function g with $\|g\|_\infty = b$ such that*

$$\mathbb{P}_\pi(S_s \leq 0) \geq 1 - \frac{4}{s}.$$

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Consequently, for every constant $c > 0$,

$$\sup_x \bigl|\mathbb{P}_\pi(S_s/c \le x) - \mathbb{P}(Z \le x)\bigr| \ge \frac{1}{2} - \frac{4}{s}.$$

The second statement follows from the first at $x = 0$, since $\mathbb{P}(Z \le 0) = 1/2$. With $c = \sigma s^{1/2}$ it shows that a function ψ for which (1.1) holds for all chains satisfies $\psi(b, s) \ge 1/2 - 4/s$ whenever $s \ge 2b$, so $\liminf_{s\to\infty} \psi(b, s) \ge 1/2$ for every $b \ge 1$; a standardized g has $\|g\|_\infty \ge 1$, so the values $b < 1$ do not arise. The chains have four states, so the answer remains negative if ψ may also depend on the number of states, and with $c = (\operatorname{var}_\pi S_s)^{1/2}$ the theorem shows that normalizing by the exact standard deviation in place of $\sigma s^{1/2}$ does not help. By the scaling in Remark 1, the theorem holds with any prescribed value of τ_2 in place of 1 and $t = s\tau_2$ in place of s.

What is missing is a lower bound for σ^2/τ_2. A Berry–Esseen inequality of Lezaud [2, Theorem 3.1] gives, for the stationary chain,

$$\sup_x \Bigl|\mathbb{P}_\pi\Bigl(\frac{S_t}{\sigma t^{1/2}} \le x\Bigr) - \mathbb{P}(Z \le x)\Bigr| \le \frac{26\,\|g\|_\infty \tau_2}{\sigma t^{1/2}} + e^{-t/\tau_2} = 26\,\|g\|_\infty \Bigl(\frac{\tau_2}{\sigma^2}\Bigr)^{1/2} \Bigl(\frac{t}{\tau_2}\Bigr)^{-1/2} + e^{-t/\tau_2}, \tag{1.2}$$

so a bound of the form (1.1) holds on any class of chains on which σ^2/τ_2 is bounded below. On two states every standardized g is an eigenfunction of Q and $\sigma^2 = 2\tau_2$, the largest value allowed by Proposition 4.29 of [1]. In the chains of Theorem 1, σ^2 is of order s^{-2} while $\tau_2 = 1$ (Remark 2).

2. The chains

Fix $b \ge 1$ and $s \ge 2b$, and put

$$\varepsilon = s^{-2}, \qquad R = s^4, \qquad \alpha = \Bigl(\frac{1 - b^2\varepsilon}{1 - \varepsilon}\Bigr)^{1/2}, \qquad m = \frac{b\varepsilon}{\alpha(1 - \varepsilon)}.$$

Let Y be the chain on $\{0, 1\}$ that jumps from 0 to 1 at rate ε and from 1 to 0 at rate $1 - \varepsilon$, and let V be the chain on $\{-1, 1\}$, independent of Y, that jumps from 1 to -1 at rate $R(1 + m)/2$ and from -1 to 1 at rate $R(1 - m)/2$. The chain of Theorem 1 is $X = (Y, V)$ on $I = \{0, 1\} \times \{-1, 1\}$, and

$$g(y, v) = \alpha(1 - y)v + by.$$

Thus $g = \pm\alpha$, with the sign of V, while $Y = 0$, and $g = b$ while $Y = 1$.

Lemma 1. (a) $3/4 \le \alpha^2 \le 1$ *and* $b\varepsilon \le m < 1/2$.

(b) *X is irreducible and reversible, and its stationary distribution is $\pi(y, v) = \pi_Y(y)\pi_V(v)$, where $\pi_Y(0) = 1 - \varepsilon$, $\pi_Y(1) = \varepsilon$, $\pi_V(1) = (1 - m)/2$ and $\pi_V(-1) = (1 + m)/2$.*

(c) *The eigenvalues of $-Q$ are 0, 1, R and $R + 1$, so $\tau_2 = 1$.*

(d) *g is standardized and $\|g\|_\infty = b$.*

Proof. (a) Since $s \ge 2b \ge 2$, we have $\varepsilon \le 1/4$ and $b^2\varepsilon = (b/s)^2 \le 1/4$. Hence $\alpha^2 \ge 1 - b^2\varepsilon \ge 3/4$, and $\alpha^2 \le 1$ because $b \ge 1$. So $\alpha(1 - \varepsilon) \le 1$, which gives $m \ge b\varepsilon$. Also $b\varepsilon \le 1/(2s) \le 1/4$ and $\alpha(1 - \varepsilon) \ge (\sqrt{3}/2)(3/4) > 3/5$, so $m < 5/12$. In particular all the rates are positive.

(b) The chains Y and V are irreducible, and they are reversible with respect to π_Y and π_V, since $(1 - \varepsilon)\varepsilon = \varepsilon(1 - \varepsilon)$ and $\frac{1-m}{2} \cdot \frac{R(1+m)}{2} = \frac{1+m}{2} \cdot \frac{R(1-m)}{2}$. A pair of independent irreducible reversible chains is an irreducible chain, reversible with respect to the product of the two stationary distributions.

(c) Put $u(y, v) = y - \varepsilon$ and $w(y, v) = v + m$. A direct computation gives $-Qu = u$ and $-Qw = Rw$; since u depends only on y and w only on v, also $-Q(uw) = (R + 1)uw$, and $-Q\mathbf{1} = 0$. The four functions $\mathbf{1}, u, w, uw$ are linearly independent, so these are all the eigenvalues, and $\lambda_2 = 1$ because $R = s^4 > 1$.

(d) Under π the coordinates are independent, $\pi(Y=1)=\varepsilon$ and $\mathbb{E}_\pi V=-m$. So $\pi(g)=-\alpha(1-\varepsilon)m+b\varepsilon=0$ by the choice of m, and, since $v^2=1$, $\pi(g^2)=\alpha^2(1-\varepsilon)+b^2\varepsilon=1$ by the choice of α. Finally $\|g\|_\infty=\max(\alpha,b)=b$. □

3. Proof of Theorem 1

Run X from π, so that Y and V are independent stationary chains. Let H be the event that $Y_u=0$ for all $u\in[0,s]$, and put $J=\int_0^s V_u\,du$. On H we have $S_s=\alpha J$ with $\alpha>0$, so

$$\mathbb{P}_\pi(S_s\le 0)\ge \mathbb{P}_\pi(H\cap\{J\le 0\})\ge \mathbb{P}_\pi(H)-\mathbb{P}_\pi(J>0). \tag{3.1}$$

First, $\mathbb{P}_\pi(H)=(1-\varepsilon)e^{-\varepsilon s}\ge(1-\varepsilon)(1-\varepsilon s)\ge 1-\varepsilon(1+s)$. Second, $\mathbb{E}_\pi J=-ms$, and since $V+m$ is an eigenfunction of the generator of V with eigenvalue $-R$, $\operatorname{cov}_\pi(V_0,V_u)=(1-m^2)e^{-R|u|}$. Hence

$$\operatorname{var}_\pi J=2\int_0^s (s-u)(1-m^2)e^{-Ru}\,du\le\frac{2(1-m^2)s}{R},$$

and Chebyshev's inequality gives

$$\mathbb{P}_\pi(J>0)=\mathbb{P}_\pi(J-\mathbb{E}_\pi J>ms)\le\frac{\operatorname{var}_\pi J}{(ms)^2}\le\frac{2(1-m^2)}{Rm^2s}.$$

By Lemma 1(a), $m\ge b\varepsilon=bs^{-2}$, so $Rm^2s\ge b^2s\ge s$ and this is at most $2/s$. Since $s\ge 2$, also $\varepsilon(1+s)=s^{-1}+s^{-2}\le 3/(2s)$. By (3.1),

$$\mathbb{P}_\pi(S_s\le 0)\ge 1-\frac{3}{2s}-\frac{2}{s}\ge 1-\frac{4}{s}.$$

Together with Lemma 1 this proves Theorem 1. □

4. Remarks

Remark 1. For $\kappa>0$ the chain $X^\kappa_t=X_{\kappa t}$ has generator κQ, relaxation time τ_2/κ and asymptotic variance rate σ^2/κ, and its integral is $S^\kappa_t=S_{\kappa t}/\kappa$. So $S^\kappa_t/(\sigma_\kappa t^{1/2})=S_{\kappa t}/(\sigma(\kappa t)^{1/2})$ with $\sigma^2_\kappa=\sigma^2/\kappa$, and the two sides of (1.1) for X^κ at time t coincide with those for X at time κt. Any value of τ_2 can therefore be prescribed in Theorem 1; equivalently, the rates may be divided by R, which gives a chain whose rates are at most 1, with $\tau_2=R=s^4$ and $t=s^5$.

Remark 2. In the eigenbasis of Lemma 1(c),

$$g=(b+\alpha m)\,u+\alpha(1-\varepsilon)\,w-\alpha\,uw,$$

the constant term vanishing by the choice of m. The functions u, w and uw are orthogonal in $L^2(\pi)$, with $\pi(u^2)=\varepsilon(1-\varepsilon)$, $\pi(w^2)=1-m^2$ and $\pi((uw)^2)=\varepsilon(1-\varepsilon)(1-m^2)$, and $\sigma^2=2\sum_k\pi(g_k^2)/\lambda_k$ over the components g_k of g in an orthogonal eigenbasis (see the proof of Proposition 4.29 in [1]). Hence

$$\sigma^2=2(b+\alpha m)^2\varepsilon(1-\varepsilon)+2\alpha^2(1-m^2)(1-\varepsilon)\Big(\frac{1-\varepsilon}{R}+\frac{\varepsilon}{R+1}\Big),$$

and $s^2\sigma^2\to 2b^2$ as $s\to\infty$ with b fixed, while $\tau_2=1$. The normalized variable $W_s=S_s/(\sigma s^{1/2})$ tends to 0 in probability. Indeed, by Chebyshev's inequality and the bound for $\operatorname{var}_\pi J$ in Section 3, $|J-\mathbb{E}_\pi J|\le ms/2$ with probability at least $1-8/s$; on this event intersected with H we have $|S_s|\le\frac{3}{2}\alpha ms$, and $\alpha ms/(\sigma s^{1/2})=b/((1-\varepsilon)\sigma s^{3/2})$, which tends to 0 at the rate $s^{-1/2}$. So the distance in (1.1) tends to $1/2$ along these chains. On the other hand $\mathbb{E}_\pi W_s^2=\operatorname{var}_\pi S_s/(s\sigma^2)\ge A(s)\to 1$ by Proposition 4.29 of [1]: the variance of W_s is carried by the complement of H, an event of probability about $1/s$ on which $|W_s|$ is of order $s^{1/2}$.

Remark 3. While $Y = 0$, the fast chain V averages $g = \pm\alpha$ to the small negative drift $-\alpha m$, and the rare and comparatively long visits of Y to 1, where $g = b$, restore the mean 0. Over a time s the chain Y makes no visit to 1 with probability $1 - O(1/s)$, and then S_s is negative with probability close to 1. For sums of n independent copies of a standardized random variable bounded by b, the classical Berry–Esseen theorem gives a bound $Cbn^{-1/2}$, because the third absolute moment is at most b. For a Markov chain the variance that governs S_t is σ^2, which Proposition 4.29 of [1] bounds above by $2\tau_2$ but which $\|g\|_\infty$ and τ_2 do not bound below; by (1.2), a lower bound for σ^2/τ_2 in terms of $\|g\|_\infty$ would give a function ψ as in (1.1).

5. Computations

The repository listed under Data availability contains a Python script, `checks.py`, with its output, and a Lean file, `Bounds.lean`. With b and s as symbols, the script verifies with SymPy the detailed balance equations, $\pi Q = 0$, the eigenfunctions of Lemma 1(c), $\pi(g) = 0$ and $\pi(g^2) = 1$, the expansion of g in Remark 2, and the formula for σ^2 there, the last through the solution h of $-Qh = g$, $\pi(h) = 0$, and $\sigma^2 = 2\pi(gh)$. For $b \in \{1, 1.5, 2, 3, 10\}$ and $s \in \{2b, 3b, 10b, 1000b\}$ it checks in 50-digit arithmetic detailed balance, the eigenfunctions, the spectrum (also with a general eigenvalue routine), the standardization of g, $\|g\|_\infty = b$ and the bounds of Lemma 1(a); it computes $\mathbb{P}_\pi(H)$ from the exponential of Q restricted to $\{Y = 0\}$ and compares it with $(1-\varepsilon)e^{-\varepsilon s}$, checks the bound for $\operatorname{var}_\pi J$ used in Section 3 and, for $s \le 6$, the formula for $\operatorname{var}_\pi J$ against numerical integration of the covariance of V computed from the matrix exponential, and computes σ^2 from a linear solve of the Poisson equation and compares it with the formula of Remark 2; for $b = 1$, where $\alpha = 1$, it checks $\pi(g) = 0$ and $\pi(g^2) = 1$ in exact rational arithmetic. It evaluates the bound $1 - \varepsilon(1+s) - 2(1-m^2)/(Rm^2 s)$ of Section 3 on a grid of b and s and checks that it is at least $1 - 4/s$, and at five pairs (b, s) with $s \le 8$ it computes $\mathbb{P}_\pi(S_s \le 0)$ by numerical Fourier inversion of $\mathbb{E}_\pi e^{i\theta S_s} = \pi^{\mathsf{T}} \exp(s(Q + i\theta G))\mathbf{1}$, where G is the diagonal matrix of the values of g, and by simulation of the chain. At $b = 2$ and $s = 8$, for example, the inversion gives 0.8844, the simulation 0.8894 with standard error 0.0044, and the bound of Section 3 0.8018. The Lean file, checked with Lean 4 [3] (version 4.33.1) and Mathlib [4], proves from the defining relations of ε, α and m the inequalities of Lemma 1(a), and it proves $(1-\varepsilon)e^{-\varepsilon s} \ge 1 - \varepsilon(1+s)$ and $\varepsilon(1+s) + 2(1-m^2)/(Rm^2 s) \le 4/s$ for $m \ge b\varepsilon$, which is the real-variable part of the proof. These computations test the identities and inequalities used in the proofs; they are not part of them.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. Open Problem 4.30 of [1] was among the open problems in probability on which the project's agents worked in September 2026. On 6 September 2026 an agent running in OpenAI's Codex found a family of four-state chains with $\|g\|_\infty = 2$ along which $S_t/(\sigma t^{1/2})$ tends to 0 in probability as $t/\tau_2 \to \infty$, which answers the question in the negative. On 10 October 2026 a second Codex agent gave a rational parametrization of these chains and showed that the distance in (1.1) tends to $1/2$ along them. On 13 October 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, found the chains of Theorem 1, which cover every value of $\|g\|_\infty$, proved the explicit bound, searched the literature, and wrote the remarks, the computer checks, the Lean file and the manuscript. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code and the Lean file are available at `https://github.com/yspennstate/relaxation-time-normal-approximation`.

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